ORF 245 – Midterm Exam

Fall 2024

- Duration of examination: 80 minutes.
- You may write your solutions using an iPad or other device, or you may write your solutions on separate paper or in an exam booklet. In either case, make sure you upload your exam onto Gradescope with your answers to the separate problems indicated.
- This exam is both open-note and open-book. You may reference class notes, the textbook, previous homework, and precept notes. You may also use a calculator.
- You may **not** use any other online or external materials.
- Collaboration is (of course) not permitted. You should not discuss the exam questions with others until grades are released and solutions have been posted.
- Unless stated otherwise, please ensure you show all important steps in deriving your final answer.
- Please be sure to clearly box your final answers to each (sub-)question.
- Please write out and sign the following pledge on the first page of your solutions:
  "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this examination."
- **Remember to upload your exam onto Gradescope and properly assign pages of your submission to each questions.**

1. **(25 points, 5 points for each part)**
   Suppose we have three events A, B, C with $P(A) = 0.3$, $P(B) = 0.4$, and $P(C) = 0.5$. Suppose we also know that $P(A \cap B) = 0.1$, $P(A \cap C) = .2$, $P(B \cap C) = 0.3$, and $P(A \cap B \cap C) = .01$. Find the probability of each of the following events:

   (a) $A \cup B$

   (b) $A^c \cap B^c$

   (c) $A \cup B \cup C$

   (d) $A^c \cap B^c \cap C^c$

   (e) $(A^c \cap B^c) \cup C$

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2. **(25 points, 5 points for each part)**
   Consider tossing a fair coin, and let X be the toss on which the first head appears. Let $Y = \min\{X, 4\}$. That is, $Y = X$ if $X = 1, 2, 3$, and $Y = 4$ otherwise.

   (a) Find the pmf of X.

   (b) Find the pmf of Y,

   (c) Sketch the cdf of Y.

   (d) Let A be the event that Y is odd, B be the event that Y is even. Are A and B independent events? Justify your answer.

   (e) Find $\mathbb{E}[Y]$.

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3. **(20 points, 5 points for each part)**
   A diagnostic test for a disease has a probability 0.9 of giving a positive result when applied to an individual who has the disease, and a probability 0.01 of giving a (false) positive when applied to someone who does not have the disease. It is estimated that 0.1% of the population have the disease. Suppose we apply the test to a random person. Calculate the following probabilities:

   (a) The test result will be positive.

   (b) Given a positive result, the individual tested has the disease.

   (c) Given a negative result, the person does not have the disease.

   (d) The probability that the person will be misclassified (i.e. the result of the test does not correspond to their state of disease).

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4. **(30 points, 5 points for each part)**
Consider the random variables X, Y with the following joint PDF:

$$f(x,y) = \begin{cases} c\,xy & 0 \le x \le 1,\, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) What value of c ensures that $f(x,y)$ is a valid joint PDF.

(b) Find the marginal PDF of X.

(c) Find $\mathbb{E}[X]$.

(d) Find the conditional density of Y given that $X = 0.5$, i.e. find $f_{Y|X}(y \mid X = 0.5)$.

(e) Find $\mathbb{E}[XY]$.

(f) Are X and Y independent? Justify your answer.